

Partial Differentiation

LLE – Mathematics and Statistics Skills

Differentiation

1. For each of the functions below, $y = f(x)$, find the derived function,
 $\frac{dy}{dx} = f'(x)$

(a) $y = 5x^3 + 10x - 8$

(b) $y = \frac{10}{x^2} - 5\sqrt{x}$

(c) $\ln 8x$

(d) $y = e^{5x} + 3e^{x^2}$

Partial Derivatives

2. For each of the functions below, $z = f(x, y)$, find:

$$\frac{\partial z}{\partial x} \quad \frac{\partial z}{\partial y} \quad \frac{\partial^2 z}{\partial x^2} \quad \frac{\partial^2 z}{\partial y^2} \quad \frac{\partial^2 z}{\partial x \partial y} \quad \frac{\partial^2 z}{\partial y \partial x}$$

(a) $z = x^2 - y^2$

(b) $z = 5x + 3x^2 - 2y^3$

(c) $z = xy$

(d) $z = x^2y^3 + 8x - 4$

(e) $z = \frac{x^2}{y^2}$

(f) $z = \ln xy$

(g) $z = e^x + 5e^{2y}$

(h) $z = x^2y - ye^x + xe^{-2y}$

(i) $z = e^{x^2-y^2}$

Implicit Differentiation

3. For each of the functions below, use implicit differentiation to find an expression for $\frac{dy}{dx}$

(a) $x^2 + 4y = 0$

(b) $x^2 - y^2 = 0$

(c) $4xy + y^2 = 0$

Total Differential

4. Find the total differential, dz , of $z = f(x, y)$ using:

$$dz = \frac{\partial z}{\partial x}dx + \frac{\partial z}{\partial y}dy$$

(a) $z = x^2 + 4y$

(b) $z = x^2 - y^2$

(c) $z = 4xy + y^2$

(d) $z = \frac{4x^2}{y^3}$

5. For each of the functions in question 4, use the total derivative of $z = f(x, y)$ to find $\frac{dy}{dx}$ for the implicit function defined by $f(x, y) = 0$.

Confirm that a to c match with the answers to question 3.